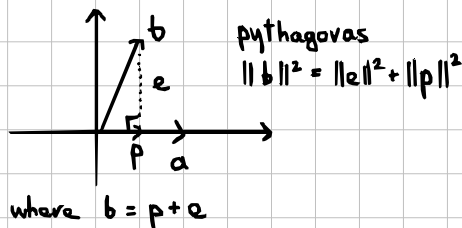


Orthogonality

- Four subspaces, two orthogonal pairs:
 - $\text{Row}(A) \perp N(A)$ in $\mathbb{R}^n$, with dimensions $r + (n-r) = n$
 - $C(A) \perp N(A^T)$ in $\mathbb{R}^m$, with dimensions $r + (m-r) = m$
- If $b \perp N(A^T)$ then b is in $C(A)$, equivalent of $Ax = b$ solvable.

Projection

	in dim 2	onto $C(A)$
solve	$\hat{x} = a^T b / a^T a$	$A^T A \hat{x} = A^T b$
point	$p = a \hat{x}$	$p = A \hat{x}$
matrix	$P = a a^T / a^T a$	$P = A (A^T A)^{-1} A^T$
check	$a^T e = 0$	$A^T e = 0$



Finance links:

- Demeaning is a projection onto 1.
- β is $a^T b / a^T a$ on demeaned returns
- $R^2 = \|p\|^2 / \|b\|^2$ (how much p explains b)

Least squares

Problem: $b \notin C(A)$, minimize $\|Ax - b\|^2$. Result is the projection:

$$A^T(A\hat{x} - b) = 0 \Leftrightarrow A^T A \hat{x} = A^T b \quad \text{with } p = A\hat{x}, e = b - p$$

Recipe:

- Form $A^T A$, $A^T b$
- Solve by elimination $\hat{x}$, don't form $(A^T A)^{-1}$
- $p = A\hat{x}$, check $A^T e = 0$

Shortcut: if $N(A^T)$ is a single line along y and $\dim N(A^T) = m - r = 1$

- $e = (y^T b / y^T y) y$ and $p = b - e$
- check $y^T p = 0$

in numpy: `np.linalg.lstsq(A, b, rcond=None)`

`np.linalg.solve(A^T A, A^T b)`

`cond(A^T A) = cond(A^2)`, big cond means unstable result.